



DECISION GOVERNANCE INFRASTRUCTURE

UNITED STATES PATENT AND TRADEMARK OFFICE

PROVISIONAL PATENT APPLICATION

TITLE OF INVENTION:

APPEND-ONLY POST-QUANTUM MERKLE TRANSPARENCY CHAIN FOR
AUTOMATED GOVERNANCE DECISION RECEIPTS WITH INTERNAL
RFC-3161-STYLE TIMESTAMPING, ROLLING HASH ACCUMULATION, AND
TAMPER-EVIDENT PUBLIC VERIFICATION

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Applicant:	OMNIX QUANTUM LTD (United Kingdom)
Entity Status:	Micro Entity — 37 C.F.R. § 1.29
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Related Applications:	OMNIX-PAT-2026-001 (Governance Control Architecture), OMNIX-PAT-2026-003 (PQC Auth)

This document constitutes a Provisional Patent Application filed pursuant to 35 U.S.C. § 111(b). It establishes a priority date and grants twelve (12) months from the filing date to submit a corresponding non-provisional application. CONFIDENTIAL — NOT FOR DISTRIBUTION.

FIELD OF THE INVENTION

The present invention relates to cryptographic audit trail systems for automated decision governance, and more particularly to an append-only transparency chain that links governance decision receipts using a rolling Merkle-style hash accumulator, applies post-quantum cryptographic signatures to each chain entry, generates RFC 3161-style trusted timestamps without dependency on external timestamp authorities, and exposes a public verification interface enabling external parties to verify the integrity and chronological ordering of the decision record without requiring access to the contents of individual receipts.

BACKGROUND OF THE INVENTION

Automated decision systems operating in regulated environments must produce audit records satisfying three independent requirements: completeness (every decision is recorded), integrity (records cannot be silently altered), and verifiability (third parties can confirm integrity without needing access to confidential contents). Existing audit trail mechanisms fail one or more of these requirements:

Log-Based Systems: Traditional logging systems record events in timestamped log files. These systems fail the integrity requirement because log files can be modified after the fact without detection — there is no cryptographic linkage between log entries that would reveal retroactive modification.

Blockchain-Based Systems: Some systems use public or permissioned blockchains as immutable ledgers. These systems require network connectivity to the blockchain, expose governance records to the blockchain's consensus and availability properties, and create regulatory uncertainty regarding the legal status of blockchain-recorded evidence.

External Timestamp Authorities (TSA): RFC 3161 timestamping requires submission of document hashes to external trusted timestamp authorities. This creates dependencies on third-party services, introduces network latency into the governance pipeline, and requires ongoing contractual relationships with external parties.

Isolated Receipt Systems: Systems that generate individual cryptographic receipts per decision — without chaining those receipts — provide per-receipt integrity but cannot detect the deletion of receipts from the record. An adversary who deletes a receipt leaves no detectable gap.

No Post-Quantum Security: All known prior art audit trail systems use classical cryptographic primitives (RSA, ECDSA, Ed25519) that are vulnerable to quantum computing attacks capable of forging signatures and breaking hash function preimage resistance.

SUMMARY OF THE INVENTION

The OMNIX Transparency Chain (OTC) provides an append-only audit trail for governance decision receipts with five key architectural properties:

Hash-Linked Chain Integrity	Each chain entry contains the hash of the previous entry (prev_log_hash). Modification of any entry changes its hash, invalidating all subsequent entries. Deletion of any entry creates a detectable hash gap.
Rolling Merkle Accumulation	A rolling Merkle root accumulates the hash of every entry with O(1) update cost. The current root is a commitment to all entries ever written. Published roots enable third-party verification without content disclosure.
Internal RFC 3161-Style Timestamping	Each entry includes a cryptographically bound Internal Timestamp Token (TST) generated without dependency on any external Timestamp Authority. The TST binds payload hash, timestamp, nonce, and policy identifier.
Post-Quantum Cryptographic Signatures	Each chain entry is signed using ML-DSA-65 (Dilithium-3, NIST FIPS 204), a lattice-based digital signature algorithm resistant to quantum computing attacks.

Content-Private Public Verification	A public verification API exposes chain entries as hash sequences only. External verifiers recompute the hash chain and Merkle root without accessing the confidential content of any governance decision receipt.
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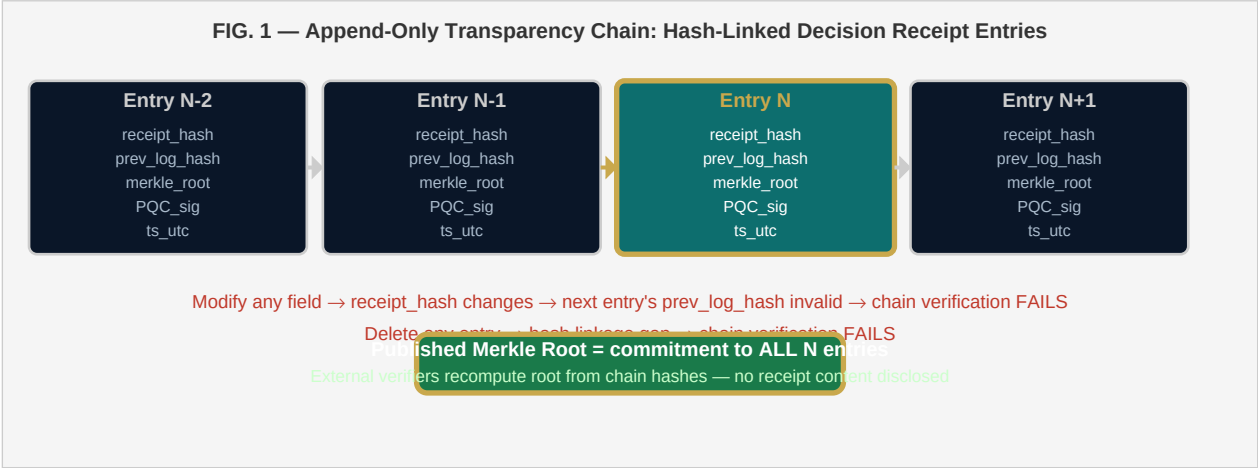


FIG. 1 — Hash-Linked Transparency Chain: Any modification or deletion detected via hash linkage failure

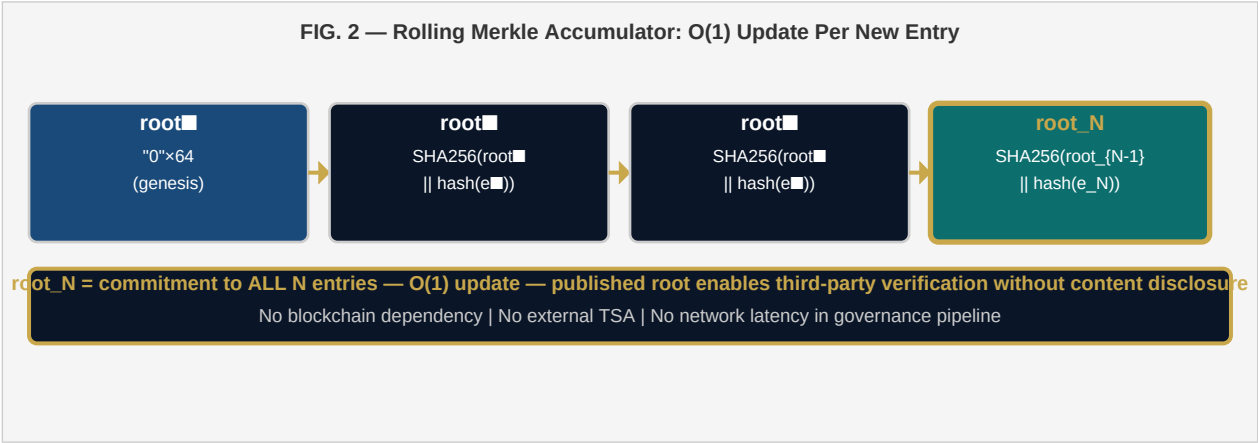


FIG. 2 — Rolling Merkle Accumulator: O(1) update per entry; current root commits to entire history

DETAILED DESCRIPTION OF THE INVENTION

I. Chain Entry Structure

Each entry in the OMNIX Transparency Chain comprises the following fields:

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receipt_hash: SHA-256 hash of the complete governance decision receipt payload |
prev_log_hash: SHA-256 hash of the previous chain entry (hash of chain entry N-1) |
merkle_root: Current rolling Merkle accumulator root after incorporating this entry | PQC_sig:
ML-DSA-65 (Dilithium-3) signature over all fields | ts_utc: Internal timestamp token (TST) in
ISO 8601 format | entry_id: Monotonically increasing entry sequence number | chain_version:
Chain format version for forward compatibility
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II. Internal Timestamp Token (TST)

The Internal Timestamp Token provides RFC 3161-equivalent trusted timestamping without dependency on external Timestamp Authorities. Each TST comprises: version (protocol version); hash_alg (hash algorithm used); payload_hash (SHA-256 of the receipt payload being timestamped); ts_utc (timestamp in ISO 8601 UTC format); nonce (16 cryptographically random bytes preventing replay); and policy (OMNIX timestamping policy identifier). The TST is then hashed: $tst_hash = \text{SHA256}(\text{canonical_json}(\text{TST}))$. This hash is included in the chain entry and signed with ML-DSA-65.



FIG. 3 — Internal Timestamp Token vs. External TSA: OMNIX eliminates network dependency and third-party risk

III. Public Verification Interface

The OMNIX Public Verification API exposes a read-only endpoint that returns chain entries as sequences of hash values — never the content of governance receipts. An external verifier can: (a) retrieve the complete chain hash sequence; (b) recompute the hash chain by verifying that each entry's hash is consistent with the previous entry's hash; (c) recompute the Merkle root to verify it matches the published root; (d) verify ML-DSA-65 signatures on each entry; and (e) verify that the hash of their own governance receipt is present in the chain — confirming their receipt was recorded and has not been modified — all without the API disclosing any governance decision content.

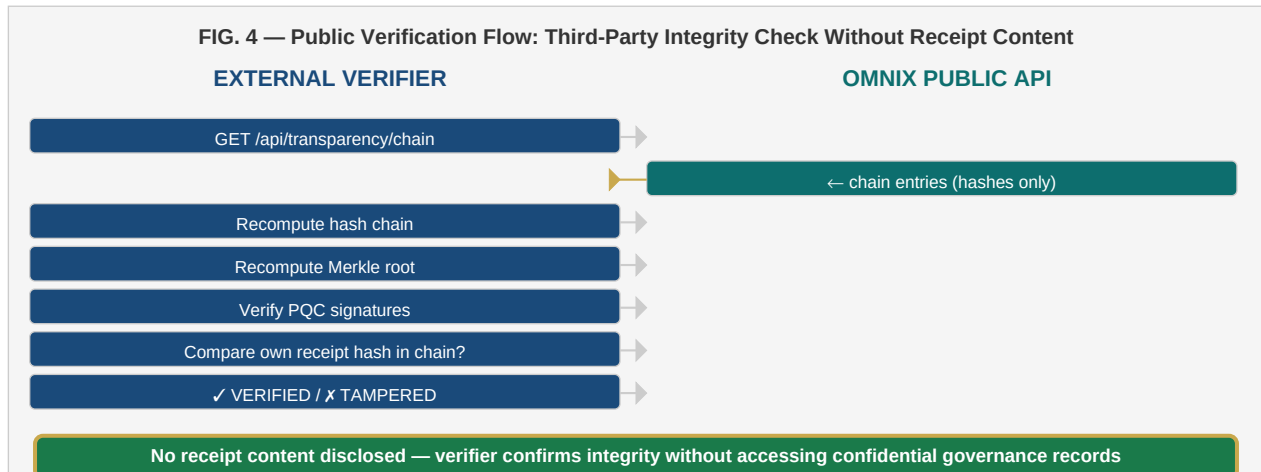


FIG. 4 — Content-Private Public Verification: Third parties verify integrity without accessing governance record contents

IV. Post-Quantum Security Properties

All chain integrity mechanisms use post-quantum cryptographic primitives. ML-DSA-65 (Dilithium-3, NIST FIPS 204) provides digital signatures resistant to attacks by quantum computers running Shor's algorithm. SHA-256 provides hash linkage and Merkle accumulation with 128-bit post-quantum security against Grover's algorithm. The complete chain integrity guarantee holds against both classical and quantum adversaries.

V. Integration with Governance Pipeline (OMNIX-PAT-2026-001)

The OMNIX Transparency Chain operates as the Layer 3 evidence and receipt layer in the four-layer governance architecture described in OMNIX-PAT-2026-001. Upon each governance decision — whether APPROVED or BLOCKED — the governance pipeline generates a decision receipt containing all checkpoint evaluations, verdicts, signal values, and the final admissibility determination. This receipt is immediately appended to the transparency chain, producing a new chain entry with a binding hash, ML-DSA-65 signature, and internal timestamp. The updated Merkle root is published to the public verification API. Observed chain append latency: 0.8–3.2 milliseconds per entry (in-memory accumulator update, asynchronous persistent write).

CLAIMS

1. A cryptographic audit trail system for automated decision governance comprising: an append-only transparency chain storing a plurality of chain entries, each chain entry comprising a receipt hash of a governance decision receipt, a previous entry hash linking each entry to the immediately preceding entry in the chain, a post-quantum digital signature over all chain entry fields, and an internal timestamp token generated without dependency on any external timestamp authority; a rolling Merkle accumulator that maintains a current Merkle root representing a commitment to all entries in the transparency chain, updated in $O(1)$ operations per new entry; and a public verification interface configured to expose chain entries as hash sequences without disclosing the content of any governance decision receipt.
2. The system of claim 1, wherein the post-quantum digital signature comprises a signature generated by the ML-DSA-65 algorithm (NIST FIPS 204, Dilithium-3) over a canonical serialization of all chain entry fields, providing signature security against adversaries equipped with quantum computing capabilities.
3. The system of claim 1, wherein the internal timestamp token comprises: a version identifier; a hash algorithm identifier; a hash of the governance decision receipt payload; a timestamp in ISO 8601 UTC format; a cryptographically random nonce of at least 16 bytes; and a policy identifier; wherein the token is internally generated without transmission to any external timestamp authority, eliminating network latency and third-party availability dependencies from the governance pipeline.
4. The system of claim 1, wherein the rolling Merkle accumulator is updated according to $\text{root_N} = \text{SHA256}(\text{root_}\{N-1\} \parallel \text{hash}(\text{entry_N}))$, wherein root_0 is a genesis value, and wherein root_N constitutes a commitment to all N entries in the transparency chain such that any modification or deletion of any entry produces a verifiably different root value.
5. The system of claim 1, wherein the public verification interface enables an external verifier to: retrieve chain entry hash sequences; recompute the hash chain and verify hash linkage integrity; recompute the Merkle accumulator root and verify it matches the published root; verify post-quantum digital signatures; and confirm that the hash of a specific governance decision receipt is present in the chain; without the interface disclosing the content of any governance decision receipt.
6. The system of claim 1, wherein modification of any chain entry causes a change in the receipt hash of that entry, which causes a mismatch in the previous entry hash field of the immediately subsequent entry, making the modification detectable through chain hash linkage verification.
7. The system of claim 1, wherein deletion of any chain entry produces a detectable gap in the hash linkage sequence, making the deletion detectable through chain hash linkage verification even without access to the deleted entry's content.
8. The system of claim 1, further configured to record both approved governance decisions and blocked governance decisions in the transparency chain, producing a complete and unbiased audit record that includes all proposed actions regardless of their admissibility determination.
9. A method for generating a tamper-evident audit trail for automated decision governance comprising: generating a governance decision receipt comprising all checkpoint evaluations, verdicts, and admissibility determination for each proposed automated action; computing a receipt hash of the governance decision receipt; constructing a chain entry comprising the receipt hash, a previous entry hash linking to the immediately preceding chain entry, a post-quantum digital signature, and an internal timestamp token; updating a rolling Merkle accumulator to produce a new Merkle root representing a commitment to all chain entries including the new entry; and publishing the updated Merkle root to a public verification interface.

10. The method of claim 9, wherein the post-quantum digital signature is generated using ML-DSA-65 (NIST FIPS 204), and wherein the internal timestamp token is generated without transmission to any external timestamp authority, such that chain entry generation does not introduce network latency or third-party service dependencies into the governance pipeline.

ABSTRACT

An append-only transparency chain for automated governance decision receipts provides cryptographically guaranteed completeness, integrity, and verifiability of the governance audit record. Each chain entry is hash-linked to the preceding entry (enabling detection of any modification or deletion), post-quantum signed using ML-DSA-65 (NIST FIPS 204), and internally timestamped using an RFC 3161-compatible Internal Timestamp Token generated without dependency on external Timestamp Authorities. A rolling Merkle accumulator maintains a current root committing to all entries with $O(1)$ update cost. A public verification interface enables external parties to verify chain integrity and confirm the presence of their governance receipts without accessing the confidential content of any receipt. The system operates as the Layer 3 evidence and receipt layer of the OMNIX four-layer governance architecture, producing a quantum-resistant, deletion-evident, and externally verifiable audit trail for all governance decisions — both approved and blocked.

Application Reference: OMNIX-PAT-2026-010 | Inventor: Harold Alberto Nunes Rodelo | OMNIX QUANTUM LTD, United Kingdom | ©
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